10/09/2025 Task-8 ~~1~~(a) 7.1(a)

Aim: write a python program to implement python generator and decorators.

Algorithm:

1) Define generator function:
2) Initialize current value:
3) Generate sequence
4) Get user input.
5) Create generator object
6) print generated sequence

Program:

```
def number_sequence
    current=start
    while current<=end
        yield current
        current+=step
start = int(input("Enter the starting number:"))
end  = int(input("Enter the ending number:"))
step = int(input)
#create the generator
sequence_generator = number_sequence
#print the generated sequence of numbers
for numbers in sequence_generator:
    print(number)
```

⇒ ~~8.1(b)~~ 7.1(b)

Aim: produce a default sequence of numbers starting from 0, ending at 10, and with a step of 1 if no values are provided.

Algorithm:

1) Start function
2) Initialize counter
3) Generate values
4) Create Generator object:
5) Iterate and print

Output:

```
Enter the starting Number = 1
Enter the ending number = 50
Enter the step value: 5
1
6
11
16
21
26
31
36
41
46.
```

Program:

```
def my-generator(n):
 #initialize counter
  value=0
 #loop until counter value of the counter
  Yield value<n:
 # produce the current value of the counter
    Yield value
 # increment the counter
     value+=1
#iterate over the generator object produced by
 my_generator
print(value)
```

Output:

Task-8 7.2

Aim: It is to work on a messaging application that needs to format messages.

Algorithm:

1) Create Decorators:
   - Define uppercase-decorator.
   - Define lowercase-decorator.
2) Define Functions:
3) Define Greet function
4) Execute the program:

program:

```
def uppercase-decorator(func):
  def wrapper(text):
   return func(text).upper()
   return wrapper
define lowercase-decorator(func):
   def wrapper(text):
   return func(text).lower()
   return wall.wrapper
@ uppercase-decorator
  def shout(text):
     return text
@ lowercase-decorator
   def whisper(text):
      return text.
def greet(func):
  greeting = func("Hi, I am function passed as an
                   argument.")
greet(shout)
greet(whisper)
```

VEL TECH	
EX NO.	7
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
WITH DATE	15

Result: Thus, the python program to implement python generator and decorators was successfully executed and the output was verified.

Output:

HI, I AM CREATED BY A FUNCTION PASSED AS AN ARGUMENT.
hi, i am created by a function passed as an argument.